

Trivanni Yadav

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Profile

Ph.D. physicist specializing in electronic-structure theory and defect thermodynamics in complex, multi-element oxide materials, using density functional theory (DFT) and climbing-image nudged elastic band (CI-NEB) calculations in VASP to compute formation energies and derive ion-migration kinetics. Complementary hands-on experience developing and validating machine-learning interatomic potentials (MLIPs) against DFT-NEB benchmarks to accelerate large-scale molecular dynamics on high-performance computing systems. This methodology is directly transferable to other complex oxide and multi-element material systems. Demonstrated record of peer-reviewed publications, national conference presentations, instructional development, and academic leadership.

Education

- **Ph.D. in Physics** (GPA: 3.82) (May 2026)
Department of Physics and Engineering Physics
The University of Tulsa, OK, USA
Advisor: Dr. Sanwu Wang | **Thesis:** A First-Principles Investigation of Defects and their Influence on Li-ion Migration in Garnet-type Solid-State Electrolyte $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$.
- **MS in Physics** (GPA: 3.34) (May 2017)
Department of Physics
Central Michigan University, MI, USA
Advisor: Dr. Christopher Tycner | **Project:** Numerical Modeling of exoplanet light curves based on Mandel and Agol analytic model.
- **M.Sc. in Physics** (Ranked Third in Class) (December 2013)
Department of Physics
Chaudhary Charan Singh University, India
- **B.Sc in Applied Physical Science** (Ranked Second in Class) (March 2012)
University of Delhi, India

Research Experience

- **Electronic-structure simulations of a complex, multi-element oxide:** Performed DFT calculations using VASP to investigate atomic and electronic properties of a garnet-type solid-state electrolyte ($\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ - four constituent elements). Analyzed charge transfer through **Bader charge analysis** and computed total and projected density of states (DOS/PDOS) to characterize the electronic structure and band gap of the pristine material.
- **Defect thermodynamics and structure-composition-property relationships:** Computed formation energies of intrinsic and complex point defects under varying **chemical potential limits**, deriving how defect stability and structure-property relationships shift across different synthesis conditions.
- **Finite-size corrections:** Applied dielectric tensor calculations and employed *sxdefectalign*, a Python-based correction scheme, to obtain accurate formation energies for charged defects.
- **Kinetic analysis via ion migration and NEB calculations:** Used the climbing-image nudged elastic band (CI-NEB) method to identify diffusion pathways and calculate migration barriers for vacancy-, interstitial-, and complex-defect-mediated ion transport – the kinetic

complement to the thermodynamic (formation-energy) analysis above.

- **Machine-learning interatomic potentials (MLIPs) for molecular dynamics (complementary project):** Implemented VASP's machine learning force field (MLFF) for on-the-fly training; conducted NVT simulations (298 – 1000 K) to extract activation energies, directly benchmarking ML-predicted migration behavior against DFT-CI-NEB results – hands-on experience validating machine-learned interatomic potentials against first-principles ground truth for accelerated, large-scale materials modeling.
- **High-performance computing workflows:** Managed parallel job execution, data handling, and automation on HPC clusters (Slurm, Titan, Schooner) for large-scale DFT and MLIP-driven molecular dynamics simulations.

Computer Skills

- **Programming & Scripting:** Python (scientific scripting, workflow automation), Bash
- **Software & Packages:** VASP, Origin, *sxdefectalign* (Python), VESTA
- **Systems:** High-Performance Computing (Slurm), GNU/Linux, LaTeX

Publications

- **T. Yadav** and S. Wang, "Migration of lithium ions and effects of defects in garnet-type solid-state electrolyte $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$," Under revision at *J. Appl. Phys.* (2026)
- **T. Yadav** and S. Wang, "Formation of defects in garnet-type solid-state electrolyte $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$," *J. Appl. Phys.* **138**, 015102 (2025) DOI: [10.1063/5.0274461](https://doi.org/10.1063/5.0274461)
- S. Liu, H. Zhang, C. Liu, S. Liu, D. Li, Z. Niu, **T. Yadav**, D. Shah and S. Wang, "Design of superhard high-entropy diborides via high-throughput DFT and thermodynamics calculations," *Ceram. Int.* **50**, 17977 (2024) DOI: [10.1016/j.ceramint.2024.02.287](https://doi.org/10.1016/j.ceramint.2024.02.287)

Conference Presentations

Oral Presentations:

- **T. Yadav** and S. Wang, "Defect-Mediated Lithium-Ion Migration and Activation Energies in Cubic $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ Solid Electrolyte," APS Global Physics Summit 2026, Denver, CO (March 17th, 2026).
- **T. Yadav** and S. Wang, "Migration of Lithium Ions and Effects of Defects in the Garnet-Type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ Solid Electrolyte," APS Global Physics Summit 2025, Anaheim, CA (March 16th, 2025).
- **T. Yadav** and S. Wang, "First-Principles Calculations of Garnet-Type Solid Electrolyte $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$," 26th Annual TU Student Research Colloquium, The University of Tulsa (March 30th, 2023).

Poster Presentations:

- **T. Yadav** and S. Wang, "Migration of Lithium Ions and Effects of Defects in the Garnet-Type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ Solid Electrolyte," Conference of Graduate Women & Gender Minorities in Physics, American Physical Society, Washington DC (September 25th, 2025).
- **T. Yadav** and S. Wang, "Migration of Lithium Ions and Effects of Defects in the Garnet-Type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ Solid Electrolyte," Graduate Research Day, The University of Tulsa (April 14th, 2025).

Oral & Poster Presentation:

- **T. Yadav** and S. Wang, “Migration of Lithium Ions and Effects of Defects in the Garnet-Type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ Solid Electrolyte,” Oxley Health & Natural Science Research Day, The University of Tulsa (April 23rd, 2025).

On demand–Virtual Presentation:

- **T. Yadav** and S. Wang, “A First-Principles Study of Defects in Garnet-Type Solid-State Electrolyte $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$,” APS March Meeting 2024, Minneapolis, MN (March 3rd, 2024).

Fellowships and Awards

- **Award for Distinction in Pedagogy** (April 16th, 2026)
The University of Tulsa
- **Dissertation Completion Fellowship** (June 2025—May 2026)
The University of Tulsa
- **Distinguished Graduate Research Poster Award** (\$500) (April 17th, 2025)
The University of Tulsa
Sole top award recipient: 1st place out of 70 graduate student participants across all academic disciplines for excellence in research presentation.
- **APS DMP Iris M. Ovshinsky Travel Award** (\$750) (March 18th, 2025)
American Physical Society (APS)
(Division of Materials Physics—DMP)
- **Graduate Summer Research Fellowship** (\$4000) (June—August, 2023)
The University of Tulsa
- **Chapman Distinguished PhD Fellowship** (June–August 2022, August–October 2024) (\$1000 each, awarded twice)
The University of Tulsa
- **Doctoral Administration Assistantship Award** (\$4800) (May—August, 2021)
The University of Tulsa
- **Graduate Teaching Assistantship Award** (August 2018—May 2025)
The University of Tulsa
- **Outstanding Teaching Assistant Award** (April 2016 & 2017)
Central Michigan University
- **Graduate Teaching Assistantship Award** (August 2014—May 2017)
Central Michigan University
- **Merit Award – First & Second Position** (2008–2009, 2010–2011, & 2011–2012)
University of Delhi

Teaching Experience

- **Instructor** (January—May 2024)
The University of Tulsa
Taught PHYS-2053 (General Physics I), a full-semester, calculus-based introduction to classical mechanics, waves, fluids, and gravity for science and engineering majors (class of 40 students). Designed and delivered lectures, developed weekly quizzes, two midterm exams, and a comprehensive final exam. Held regular office hours throughout the semester to support student learning and exam preparation.

- **Course Builder** (May 2021—Present)
The University of Tulsa
Developed online supplemental instructional resources for PHYS-2053, featuring topic videos, worked examples and self-assessment quizzes accessed by more than 300 undergraduates. Designed materials to support conceptual understanding and independent learning in introductory physics.
- **Graduate Teaching Assistant** (August 2018—December 2023, August 2024—May 2025)
The University of Tulsa
Independently taught the undergraduate General Physics I Laboratory (PHYS-2051) and the Introductory Physics Laboratory (PHYS-1011) as the sole instructor for assigned sections, serving approximately 50 students per semester. Responsibilities included delivering pre-laboratory instruction, supervising experiments, assessing laboratory reports, maintaining laboratory instrumentation, and providing student mentoring through regular office hours.
- **Graduate Teaching Assistant** (August 2014—May 2017)
Central Michigan University
Taught PHY 170/175 (24 students per class), algebra- and calculus-based physics labs; graded assignments; maintained Blackboard updates; and held weekly office hours. Coordinated experiments and setup for lab courses. Supported active-learning classes such as PHS 265 QR—Quantitative Physical Science, an undergraduate course for students preparing to teach in elementary school; graded assignments and set up experiments for the entire class.

Leadership and Academic Service

- **APS Student Ambassador** (April 2025—April 2026)
American Physical Society
- **Invited Participant**, APS Annual Leadership Meeting; (February 3rd—6th, 2026)
Selected Participant, APS Congressional Visit Day
Capitol Hill, Washington, DC
- **President**, Physics Graduate Student Association (January 2025—May 2026)
The University of Tulsa
- **Graduate Student Representative** (January 2025—May 2026)
Academic College Committee, The University of Tulsa
- **Graduate Senator**, Graduate School Association (August 2023—August 2025)
The University of Tulsa
Served on Academic and Professional Development Committee and Travel Grants Committee

Outreach and Media

- **Featured by University of Tulsa press release:** [Physics doctoral student pursues battery research in supportive community at UTulsa](#)
- **Featured on PulseraNews article:** [Doctoral Candidate Trivanni Yadav Advances Battery Research at UTulsa](#)
- **Featured on The Indiana Journal:** [Doctoral Student Innovates Battery Research at The University of Tulsa](#)

Workshops

In-Person Workshop

- **2022 summer school on electron-phonon physics from the first principles** (June 13th—20th, 2022)
(Selected Participant to attend a competitive, NSF-sponsored, Summer School on Electron-Phonon Physics from First Principles at the University of Texas at Austin, gaining hands-on experience with advanced first-principles computational methods)

Virtual Workshops

- **Machine Learning in Materials Science-From Basic Concepts to Active Learning** (January 16th, 2025)
(Organized by Materials Research Society)
- **Exploring chemical reactions in VASP** (November 6th—8th, 2024)
(Organized by VASP Software GmbH)
- **Advanced optics and dielectric properties** (March 13th—15th, 2024)
(Organized by VASP Software GmbH)
- **Python Tools for Structure-Informed Prediction of Formation Energy using Neural Networks** (March 2nd, 2023)
(Organized by Materials Genome Foundation)
- **Electronic correlation from first principles** (November 15th—18th, 2022)
(Organized by VASP Software GmbH)
- **Moving ions with VASP** (September 27th—30th, 2022)
(Organized by VASP Software GmbH)

Professional Societies

- **Sigma Pi Sigma (Society of Physics Students)** (2017—Present)
Inducted Member, Central Michigan University Chapter.
- **American Physical Society** (2023—Present)

References

- **Dr. Sanwu Wang**
Chair, Department of Physics and Engineering Physics
The University of Tulsa, Tulsa, OK 74104
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Relationship: Ph.D. Research Advisor and Primary Supervisor
- **Dr. Scott Holmstrom**
Dean of the Graduate School, Professor of Physics & Engineering Physics
The University of Tulsa
Phone: (918) 631-3031 | Email: scott-holmstrom@utulsa.edu
Relationship: Ph.D. Dissertation Committee Member
- **Dr. Siamack Shirazi**
Professor of Mechanical Engineering
Director, Erosion/Corrosion Research Center (E/CRC) and Tulsa University Sand Management Projects (TUSMP), The University of Tulsa
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Relationship: Ph.D. Dissertation Committee Member